

Measurability of the canonical coefficients in the parameter

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Abstract

Astra #10 rank 3, completing the parameter-integration adapter. For a coefficient family jointly measurable in (w, s) with a common radius, the canonical coefficient functions $U_\alpha, V_\alpha, C_\alpha$ are jointly measurable (via the series formulas of unit 125, as pointwise-convergent series of measurable terms), log moments of jointly measurable functions are measurable in the parameter (a parametric Bochner integral), hence $w \mapsto A_\alpha(w)$, $w \mapsto B_\alpha(w)$ are measurable (`canonA_param_measurable`, `canonB_param_measurable`). Consequently the averaged and expected Taylor trees hold with joint measurability of the family as the ONLY measurability hypothesis (`twoD_taylor_tree_integrated'`, `twoD_taylor_tree_expectation'`). Zero sorry/axiom.

1 The things formalised

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theorem measurable_tsum_of_summable {X : Type*} [MeasurableSpace X] (term :  $\alpha \rightarrow X \rightarrow \mathbb{R}$ )
  (hm :  $\alpha \rightarrow j, \text{Measurable (term j)}$ ) (hs :  $\alpha \rightarrow x, \text{Summable fun j => term j x}$ ) :
  Measurable fun x =>  $\sum_j \text{term j x}$ 
theorem logMoment_param_measurable { $\Omega$  : Type*} [MeasurableSpace  $\Omega$ ] (  $\alpha : \Omega \rightarrow \mathbb{R}$  ) (  $\beta : \Omega \rightarrow \mathbb{R}$  )
  (f :  $\Omega \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ ) (hf : Measurable fun p :  $\Omega \times \mathbb{R} \Rightarrow f p.1 p.2$ ) :
  Measurable fun w => logMoment  $\alpha$  (f w)
theorem canonU_param_measurable ... (  $\alpha : \Omega \rightarrow \mathbb{R}$  ) :
  Measurable fun p :  $\Omega \times \mathbb{R} \Rightarrow$ 
    canonU b h h k k (anaFaceU (c p.1) b) (fun i j s => c p.1 (i, j) s) p.2
theorem canonA_param_measurable { $\Omega$  : Type*} [MeasurableSpace  $\Omega$ ] (  $\alpha : \Omega \rightarrow \mathbb{R}$  ) (h h k k :  $\Omega \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ )
  (c :  $\Omega \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ ) (hcm :  $\alpha \rightarrow ij, \text{Measurable fun p : } \Omega \times \mathbb{R} \Rightarrow c p.1 ij p.2$ )
  (  $\beta : \Omega \rightarrow \mathbb{R}$  ) :
  Measurable fun w => canonA  $\alpha$  h h k k (fun i j s => c w (i, j) s)
theorem canonB_param_measurable ... (  $\alpha : \Omega \rightarrow \mathbb{R}$  ) :
  Measurable fun w => canonB  $\alpha$  b h h k k (anaFaceU (c w) b) (anaFaceV (c w) b)
  (fun i j s => c w (i, j) s)
theorem twoD_taylor_tree_integrated' ...
  (hcm :  $\alpha \rightarrow ij, \text{Measurable fun p : } \Omega \times \mathbb{R} \Rightarrow c p.1 ij p.2$ ) ... : |  $Z_w(N) - \sum_{|j| \leq 2T} N^{-|j|}$ 
  }((  $A_{\alpha}$  ) log N +  $B_{\alpha}$  )|
  uniformTreeConst ... * (N ^ (-(2 * T))) * (1 + Real.log N)) *
theorem twoD_taylor_tree_expectation' ... [IsProbabilityMeasure  $\mu$ ] ... :
  |E[ $Z_N$ ] -  $\sum_{|j| \leq 2T} N^{-|j|} E[A_j]$  log N + E[ $B_j$ ]|
  uniformTreeConst ... * (N ^ (-(2 * T))) * (1 + Real.log N))
```

2 Mathematics

At a u -pole $\alpha = \alpha_i$ the canonical face coefficient is $k_1^{-1} \sum_j c_{ij}(w, s) q_{\gamma_i}(j)$ (unit 125), a series of jointly measurable terms which converges for every (w, s) by the weighted row bound; off the u -poles it van-

ishes. The collision coefficient is $c_{ij}/(k_1 k_2)$ or zero. The log moment $\int_0^\infty s^{\alpha-1} (\log s)^\ell e^{-\beta s^2} f(w, s) ds$ of a jointly measurable f is measurable in w (Fubini-type measurability of parametric integrals over the σ -finite measure on $(0, \infty)$). Sums and differences preserve measurability.

3 Paper-facing meaning

Together with units 135, 137 and 138: *the Taylor tree is uniform on bounded analytic families and commutes with finite-mass averaging*, now as a theorem whose only hypotheses are analytic (common envelope) and the natural joint measurability of the family. In the SLT application this covers both the per-stratum integral over the noncritical coordinates and the expectation over the empirical fluctuation ξ_n , with expected canonical coefficients. As Astra #10 cautions, this says nothing yet about $E[\log Z]$.

4 Lean notes

- To use a lemma stated at `uExp h k i` for a general pole with `hP : uExp (uIdx ) = ,` substitute with `obtain i, rfl : i, uExp h k i = := _, hP ;` rewriting `← hP` would loop through `uIdx` .
- Anonymous-function arguments whose binder type cannot be inferred from the expected type (`fun j p => c p.1 ...`) need an explicit annotation (`p : Ω ×`).
- `StronglyMeasurable.integral_prod_right'` with `( := volume.restrict (Ioi 0))` matches the set-integral notation of `logMoment` directly.